

Power Analysis using G*Power

What is statistical power?

		Study finding	
		Effect present	Effect absent
State of the world	Effect present	Reject H_0	Type II error (miss)
	Effect absent	Type I error (false alarm)	Retain H_0

Desired outcomes:

- Probability of type I error (α) less than 5% ($p \leq .05$)
- Probability of Type II error (β) less than 20% ($1 - \beta \geq 80\%$)

Effect size

- Alpha (i.e., p value) only states *whether* an effect is statistically significant
- Magnitude of effect in sample is also important
 - With enough data, 1% improvement/difference may be significant
 - With too little data, a 50% improvement/difference may be non-significant
- If p is greater than .05, effect size is still informative

Underpowered study consequences

- Cannot know whether
 - Effect does not exist
 - OR
 - Effect exist, but study was underpowered (i.e., not enough data to detect it)
- If study is conducted, participants are put at risk without hope of valid conclusion or benefits
- A potentially effective treatment may be overlooked

Required sample size is determined by 3 inputs

- Chosen alpha
- Chosen power
- Expected effect size

What effect size to expect

- Review previous literature studying similar effects and note reported effect sizes
 - Cohen's d or Hedges's g

- Eta (η , η^2)
- Omega (ω , ω^2)
- Pearson's r
- If none of the above, note means and SDs
- Statistical tests used in previous lit does not have to be identical to those appropriate to your study's design

Cohen (1992) Table 1, p.157, lists a number of effect sizes, for interpretation purposes.

Table 1
ES Indexes and Their Values for Small, Medium, and Large Effects

Test	ES index	Effect size		
		Small	Medium	Large
1. m_A vs. m_B for independent means	$d = \frac{m_A - m_B}{\sigma}$	.20	.50	.80
2. Significance of product-moment r	r	.10	.30	.50
3. r_A vs. r_B for independent r s	$q = z_A - z_B$ where z = Fisher's z	.10	.30	.50
4. $P = .5$ and the sign test	$g = P - .50$	.05	.15	.25
5. P_A vs. P_B for independent proportions	$h = \phi_A - \phi_B$ where ϕ = arcsine transformation	.20	.50	.80
6. Chi-square for goodness of fit and contingency	$w = \sqrt{\sum_{i=1}^k \frac{(P_{ii} - P_{oi})^2}{P_{oi}}}$	.10	.30	.50
7. One-way analysis of variance	$f = \frac{\sigma_m}{\sigma}$	.10	.25	.40
8. Multiple and multiple partial correlation	$f^2 = \frac{R^2}{1 - R^2}$	.02	.15	.35

Note. ES = population effect size.

Download G*Power: <https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower>

Case studies using G*Power

- Independent sample t test (e.g., treatment vs placebo group)
- Pearson's R correlation test
- Multiple Regression

References:

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